Superstore Dataset Analysis Examples

1- Frequency Table (Absolute / Raw Counts)

Q: How many orders did each region receive?
س: كم عدد الطلبات التي استقبلتها كل منطقة؟
Tool: Count each region's orders in a frequency table. Look for which region has the most orders — does high volume mean high profit?

2- Frequency Table (Relative / Percentages)

Q: What share of total orders does each shipping mode represent?
س: ما النسبة المئوية التي تمثلها كل طريقة شحن من إجمالي الطلبات؟
Tool: Divide each count by 9,994 to get percentages. Look for whether one shipping mode dominates — Standard Class should be the majority.

3- Frequency Table (Cumulative)

Q: If we ranked shipping modes from most to least used, what percentage of orders do the top two modes cover together?
س: إذا رتّبنا طرق الشحن من الأكثر استخداماً إلى الأقل، فما النسبة التي تغطيها أعلى طريقتين معاً؟
Tool: Add relative frequencies progressively down the table. Look for whether two modes alone cover 75–80% of all orders, meaning the others are rarely used.

4- Cross-tabulation Table

Q: Do Corporate customers use expedited shipping more than Consumer customers?
س: هل يستخدم عملاء الشركات الشحن السريع أكثر من العملاء الأفراد؟
Tool: Build a table with Segment as rows and Ship Mode as columns, counts in the cells. Look for whether Corporate rows show higher proportions in First Class and Same Day columns compared to Consumer rows.

5- Bar Chart

Q: Which product category receives the most orders?
س: أي فئة منتجات تستقبل أكبر عدد من الطلبات؟
Tool: Bar chart with Category on the horizontal axis and order count on the vertical axis. Look for the tallest bar — and whether high order count matches high profit or not.

6- Side-by-Side Bar Chart

Q: Does the West region outperform other regions in sales across all customer segments, or only in some?

س: هل تتفوق منطقة الغرب على بقية المناطق في المبيعات عبر جميع شرائح العملاء، أم في بعضها فقط؟
Tool: Clustered bar chart with Region on the horizontal axis, one bar per Segment within each cluster. Look for whether one region-segment combination clearly dominates, and whether the pattern is consistent across all regions or varies.

7- Pie Chart

Q: What share of total revenue does each product category contribute?
س: ما حصة كل فئة منتجات من إجمالي الإيرادات؟
Tool: Pie chart with each slice representing a category's percentage of total Sales. Look for whether one category dominates the revenue mix — and whether that dominance matches its share of orders or profit.

8- Pareto Chart

Q: Which sub-categories should we prioritize in inventory investment because they drive the bulk of our sales?
س: أي الفئات الفرعية يجب أن نُولِيها الأولوية في الاستثمار لأنها تولّد الجزء الأكبر من مبيعاتنا؟
Tool: Bar chart sorted highest to lowest Sales, with a cumulative percentage line overlaid. Look for the point where the cumulative line crosses 80% — the bars to the left of that point are your vital few sub-categories worth focusing on.

9- Histogram

Q: Are most orders modestly profitable, or is profit spread evenly across a wide range?
س: هل معظم الطلبات ذات ربحية متواضعة، أم أن الأرباح موزّعة بالتساوي عبر نطاق واسع؟
Tool: Histogram of the Profit column, with profit ranges on the horizontal axis and order count on the vertical axis. Look for where the tallest bars sit, how wide the spread is, whether the shape is symmetric or skewed, and whether outliers appear.

10- Scatter Plot

Q: Does giving a higher discount tend to reduce profit, and how consistent is that pattern?
س: هل تميل الخصومات الأعلى إلى تقليل الربح، وما مدى اتساق هذا النمط؟
Tool: Scatter plot with Discount on the horizontal axis and Profit on the vertical axis, with a linear trendline added. Look for direction, clustering, and whether high discounts fall below zero profit.

11- Boxplot

Q: Which product category has the most unpredictable profit, and which has the most loss-making outlier orders?

س: أي فئة منتجات تمتلك أكثر الأرباح غير القابلة للتنبؤ، وأيها يحتوي على أكبر عدد من الطلبات الخاسرة الشاذة؟

Tool: Side-by-side boxplot per category. Look at IQR width, median position, whiskers, and outliers.

12- Mean

Q: On average, how much profit does a Superstore order generate?

س: كم يبلغ متوسط الربح الذي يحققه طلب واحد في المتجر؟

Tool: =AVERAGE() on Profit. Check sign and sensitivity to outliers.

13- Median

Q: What profit does the middle-ranked order earn?

س: كم يبلغ ربح الطلب المتوسط؟

Tool: =MEDIAN() on Profit. Compare to mean for skewness.

14- Mode

Q: What is the most commonly applied discount rate?

س: ما معدل الخصم الأكثر شيوعاً؟

Tool: =MODE() on Discount. Identify dominant policy rate.

15- Range

Q: What is the full span of profit outcomes?

س: ما المدى الكامل لنتائج الربح؟

Tool: =MAX() − =MIN() on Profit. Note sensitivity to outliers.

16- **Mean Absolute Deviation (MAD)**

Q: On average, how far does a typical order's profit deviate from the mean — in plain dollar terms?

س: بالدولار، كم يبلغ متوسط انحراف ربح الطلب النموذجي عن المتوسط الحسابي؟

Tool: =AVEDEV() on the Profit column. Look for a large MAD meaning orders are routinely far from average profit, confirming high variability.

17- **Variance**

Q: How do we mathematically quantify the total spread of profit outcomes across all orders?

س: كيف نقيس رياضياً التشتت الكلي لنتائج الأرباح عبر جميع الطلبات؟

Tool: =VAR.S() on the Profit column. Look for a high value confirming high dispersion (in squared units).

18- **Standard Deviation**

Q: In dollar terms, how much does a typical order's profit differ from the average?

س: بالدولار، كم يختلف ربح الطلب النموذجي عن المتوسط؟

Tool: =STDEV.S() on the Profit column. Look for SD larger than the mean — a warning sign of instability.

19- **Coefficient of Variation (CV)**

Q: Which category is most financially risky relative to its average profit?

س: أي الفئات أعلى مخاطرة مالية مقارنة بمتوسط أرباحها؟

Tool: CV = Standard Deviation ÷ Mean (per category). Look for the highest CV as the most volatile category.

20- **Skewness**

Q: Is profit distribution skewed toward large gains or losses?

س: هل توزيع الأرباح منحاز نحو مكاسب كبيرة أم خسائر كبيرة؟

Tool: =SKEW() on Profit. Positive = right tail (big gains), negative = left tail (loss-heavy).

21- **Percentiles & Quartiles**

Q: What profit levels define the bottom 25%, middle, and top 10% of orders?

س: ما مستويات الربح التي تحدد أدنى 25% والنطاق الأوسط وأعلى 10%؟

Tool: =QUARTILE.INC() and =PERCENTILE.INC(). Check if Q1 is negative and IQR width.

22- **IQR & Outlier Fences**

Q: Which orders are extreme outliers in profit or loss?

س: أي الطلبات تعتبر قيماً شاذة جداً في الربح أو الخسارة؟

Tool: IQR = Q3 − Q1; fences = Q1 − 1.5×IQR and Q3 + 1.5×IQR. Use =COUNTIF() to detect outliers.

23- **Covariance**

Q: Do Sales and Profit move together across orders?

س: هل تتحرك المبيعات والأرباح معاً عبر الطلبات؟

Tool: =COVARIANCE.S() on Sales and Profit. Positive = move together (scale-dependent).

24- **Correlation**

Q: How strong is the relationship between Discount and Profit (and Sales vs Profit)?

س: ما قوة العلاقة بين الخصم والربح (وكذلك بين المبيعات والربح)؟

Tool: =CORREL(). Interpret sign and magnitude (≥0.7 strong, 0.4–0.7 moderate, <0.4 weak).

25- **Z-Score**
Q: Is a $500 profit order exceptional or normal?
س: هل ربح 500 دولار يعتبر استثنائياً أم طبيعياً؟
Tool: Z = (X − Mean) ÷ Standard Deviation. Look for > +2 (very high) or < −2 (extreme loss).

26- **Empirical Rule**
Q: Where should ~95% of Sales values fall if distribution is normal?
س: بين أي قيمتين تقع تقريباً 95% من المبيعات إذا كان التوزيع طبيعياً؟
Tool: Mean ± 2×Standard Deviation. Check if bounds are realistic.

27- **Normal Distribution Probability**
Q: What proportion of orders are expected to lose money, and what defines top 10% profit?
س: ما نسبة الطلبات المتوقع أن تخسر، وما حد أفضل 10% من الأرباح؟
Tool: =NORM.DIST() and =NORM.INV(). Compare model vs actual distribution.

28- **Confidence Interval**
Q: How confident are we that the sample mean profit reflects true profitability?
س: ما مدى ثقتنا بأن متوسط العينة يعكس الربحية الحقيقية؟
Tool: Mean ± CONFIDENCE.NORM() or CONFIDENCE.T(). Observe interval width vs confidence level.

29- **One-Sample t-Test**
Q: Is average profit significantly greater than zero?
س: هل متوسط الربح أكبر من الصفر إحصائياً؟
Tool: One-sample t-test vs 0. Look for $p < 0.05$ to confirm profitability.

30- **Two-Sample t-Test**
Q: Do Corporate customers spend more per order than Consumers?
س: هل ينفق عملاء الشركات أكثر من العملاء الأفراد لكل طلب؟
Tool: Two-sample t-test (unequal variances). Check p-value and practical difference.

31- **Paired t-Test**
Q: Do discounted orders meet the $50 profit target?
س: هل تحقق الطلبات المخفّضة هدف الربح البالغ 50 دولاراً؟

Tool: Compare mean of discounted orders vs $50 benchmark. Look for $p < 0.05$ (failure if lower).

32- **One-Way ANOVA**
Q: Do regions differ in average profit?
س: هل تختلف المناطق في متوسط الربح؟
Tool: ANOVA (Single Factor). If $p < 0.05$, at least one region differs.

33- **Chi-Square Test**
Q: Is there a relationship between customer segment and shipping mode?
س: هل توجد علاقة بين شريحة العميل وطريقة الشحن؟
Tool: =CHISQ.TEST(). $p < 0.05$ indicates dependency between variables.

34- **Simple Linear Regression**
Q: How much does profit change for each 10% increase in discount?
س: كم يتغير الربح لكل زيادة 10% في الخصم؟
Tool: Regression (Profit vs Discount). Interpret coefficient, p-value, and R^2.

35- **Multiple Linear Regression**
Q: How well can we predict profit using Sales and Discount, and which matters more?
س: ما دقة التنبؤ بالربح باستخدام المبيعات والخصم، وأيهما أكثر تأثيراً؟
Tool: Regression (Profit vs Sales + Discount). Compare coefficients, R^2, and significance.